

SHAHIN RAHBARIASL

SENIOR SOFTWARE ENGINEER · DISTRIBUTED SYSTEMS · ADTECH

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600B+

REQUESTS / DAY

~1.7T

DOWNSTREAM / DAY

<10ms

END-TO-END LATENCY

12

GLOBAL DATA CENTERS

SUMMARY

Senior Software Engineer with **9+ years** designing and scaling distributed systems in mission-critical production environments. Deep expertise in **Go**, architecting ad-exchange services that process **~600B requests/day** at **sub-10ms latency** across **12 global data centers** on Index Exchange's own cloud infrastructure. Currently leading a company-wide rebuild of the **authorization & identity platform** (OPA/Rego, Keycloak, Kubernetes). Proven track record in AdTech infrastructure, real-time data pipelines, and open-source standards. M.Sc. Computer Science, University of Waterloo — published researcher at SIGIR.

EXPERIENCE

Index Exchange — world's largest independent ad exchange

Mar 2023 - Present

Senior Software Engineer

- Architect core ad-exchange services processing **~600B requests/day**, fanned out to **100+ demand partners** generating **~1.7T downstream requests daily**.
- Maintain **sub-10ms end-to-end latency** across thousands of concurrent Go processes on **12 globally distributed data centers** — auction decisions made in single-digit milliseconds.
- Lead the ground-up rebuild of the **company-wide authorization & identity platform** — one of the most critical infrastructure programs — with **OPA/Rego policy-as-code**, **Keycloak**, **GraphQL/Hasura**, **Kubernetes**, **ArgoCD**, and **Go**.
- Design **per-account, fine-grained access control** and the new **User API** and accounts-data services behind it, including the data migrations to move the org onto the new model safely.
- Build internal developer tooling — including **AI-assisted code review** — to raise quality and velocity across a large engineering organization.
- Build **Privacy Sandbox** exchange components and header-bidding solutions across **Web, CTV, and Mobile**; open-source contributor to **Prebid** and **Adscert** (IAB Tech Lab security protocol).

Stack: Go · Kubernetes · ArgoCD · OPA/Rego · Keycloak · GraphQL · Hasura · Kafka · Aerospike · gRPC · private cloud

Index Exchange

May 2021 - Mar 2023

Software Engineer II

- Designed high-performance Go services and APIs for the core ad-exchange platform with a distributed, low-latency architecture built to run hot under real-time load.
- Built the storage layer (**Aerospike**, **MariaDB**, **MySQL**), streaming pipelines (**Kafka**), and full observability stack (**Grafana**, **InfluxDB**).
- Automated CI/CD workflows with Ansible and GitHub/GitLab across large-scale global infrastructure.

Seneca Polytechnic — Software Design & Data Science

Jan 2024 - Apr 2024

Faculty (Part-time)

- Developed and delivered the **Web Programming Principles** curriculum; mentored students on modern frameworks and tech-career development.

IBM — Cloud & Cognitive Software

Mar 2020 - May 2021

Data Engineer

- Built core IBM Cloud services: **authentication controllers**, policy management, and the Commerce back-end serving thousands of enterprise customers.
- Architected **telemetry data pipelines** for cloud-infrastructure monitoring at scale during the pandemic-era digital-transformation surge.

Blockthrough

Jul 2018 - Mar 2020

Full Stack Software Engineer

- Joined as **engineer #10** and scaled with the team to **50+ engineers** — owned data-pipeline architecture, real-time monitoring, and backend infrastructure.
- Built **Bidthru** (Prebid wrapper) processing **~1B auctions/day**, generating **\$1M+ revenue** in its first iteration.
- Developed publisher analytics dashboards and animation-detection tooling for ad-inventory quality assurance.

BMO Capital Markets

May 2017 - Sep 2017

Software Developer (Internship)

- Built a real-time client/trader analytics platform used by trading-floor managers for P&L and performance tracking.
- Optimized financial databases and implemented live trade monitoring for risk management and compliance (CVA/FVA/XVA aggregation).

Research & Teaching Assistant

- Conducted research in information retrieval and HCI under Dr. Mark D. Smucker; led user studies and built evaluation tooling — **published 2 papers at SIGIR 2019**.
- Teaching Assistant for Operating Systems and Introduction to Computer Science.

SELECTED RECENT INITIATIVES · INDEX EXCHANGE

Authorization & Identity Platform

Per-account, policy-as-code access control built on OPA/Rego and Keycloak, replacing legacy authorization org-wide.

AI-Assisted Code Review

Internal tooling that augments code review to raise quality and velocity across a large engineering org.

Observability & Alerting

Grafana alerting and end-to-end flow validation for early detection across the request path.

User API & Accounts-Data Services

New account-domain services and APIs backing the platform, with zero-downtime data migrations onto the new model.

Feature Toggles & Safe Rollouts

Feature-flag infrastructure enabling controlled, progressive delivery across global services.

Go Platform Libraries & Proto Tooling

Shared libraries for signal management, inventory grouping, and protobuf/JSON tooling across the monorepo.

TECHNICAL SKILLS

Languages

Go · Python · Java · TypeScript · JavaScript · C++ · C# · Bash

Cloud & DevOps

Kubernetes · Docker · ArgoCD · Ansible · AWS (S3, Lambda, Kinesis, Athena, Glue) · GCP

Web & APIs

gRPC · GraphQL · Hasura · REST · React · Node.js · Angular

Systems & Data

Kafka · Spark · Redis · Cassandra · Aerospike · PostgreSQL · MySQL · MongoDB · Elasticsearch

Authz & Identity

OPA / Rego · Keycloak · policy-as-code · RBAC · Vault

Observability

Grafana · Prometheus · InfluxDB · Zookeeper

SELECTED PUBLICATIONS

SIGIR '19 Time-Limits and Summaries for Faster Relevance Assessing

SIGIR '19 Efficient Construction of Test Collections Using Dynamic Sampling

THESIS The Effects of Time Constraints & Document Excerpts on Relevance Assessing Behaviour — M.Sc. Thesis, U. Waterloo

BOOK '15 Web Development Using Drupal CMS (ISBN 9789648340419)

OPEN SOURCE

Adscert (IAB Tech Lab, 2021) — cryptographic authentication protocol for securing programmatic ad transactions; now an industry standard.

Prebid (2021) — leading open-source header-bidding library enabling publishers worldwide to maximize ad revenue.

EDUCATION

M.Sc. Computer Science

University of Waterloo · 2016–2018

B.Sc. Computer Science

Iran Univ. of Science & Technology · 2012–2016

AWARDS

- ★ **Employee of the Year**, Blockthrough · 2018–19
- ★ **International Masters Student Award**, U. Waterloo · 2016
- ★ **Graduate Research Studentship**, U. Waterloo · 2016
- ★ **Top 1%** — Iran National University Entrance Exam (300,000+ participants) · 2012
- ★ **Full Scholarship**, Iran Univ. of Science & Technology · 2012